AHMET ULU

Electrical & Electronics Engineering Student | Embedded Systems | IoT | MCU Developer

Kayseri, Türkiye | ahulu2006@gmail.com | linkedin.com/in/ahmet-ulu-16099832b | ahmetulu.com.tr

PROFILE

Electrical and Electronics Engineering student at Abdullah Gül University with a dedicated focus on embedded systems, IoT architectures, and hardware-software co-design. Hands-on experience developing integrated applications using microcontrollers (ESP32, STM32, Arduino) and single-board computers (Raspberry Pi) combined with diverse sensor networks. Proven background in schematic capture and PCB layout with KiCad, alongside sensor telemetry, database connectivity, and camera-based vision pipelines. Eager to contribute practical engineering and problem-solving skills within innovative multidisciplinary engineering teams.

EXPERIENCE

Embedded Systems Developer
February 2026 – Present

MUSTECH — Kayseri, Türkiye

- Engineered embedded firmware and hardware interfaces for IoT-based environmental monitoring systems.
- Designed low-noise sensor interfacing circuits and integrated wireless telemetry protocols for reliable field data acquisition.
- Built end-to-end data pipelines connecting distributed sensor nodes with cloud databases for real-time telemetry and logging.

Freelance PCB Designer
2025 – Present

Bionluk (Freelance Marketplace) — Remote

- Designed and sold custom PCB layouts for clients through the Bionluk freelance platform, managing the full workflow from schematic capture to production-ready fabrication files.
- Handled client requirement gathering, design iteration, and delivery for multiple independent PCB design projects.

EDUCATION

Bachelor of Science in Electrical and Electronics Engineering
2024 – 2029 (Expected)

Abdullah Gül University — Kayseri, Türkiye

- Core focus areas: Circuit Analysis, Embedded Systems, IoT Architectures, Sensor Networks, Microcontroller Programming.
- Actively involved in laboratory prototyping, academic project tracks, and extracurricular engineering challenges.

Intensive English Language School Program (CEFR B2 Level)
June 2025 – August 2025

English Path — Dubai, UAE

- Completed an intensive two-month international language immersion program focusing on technical communication, academic writing, and cross-cultural teamwork.

TECHNICAL PROJECTS

- IoT Sensor Telemetry & Cloud Logging System: Developed an end-to-end telemetry station acquiring multi-sensor data, packetizing metrics via standard serial/wireless protocols, and dispatching payload to remote databases and visualization dashboards.
- Embedded Camera Recognition Pipeline: Deployed a camera module on a Raspberry Pi single-board computer running edge image processing routines for automated recognition and local telemetry logging.
- Custom Hardware & PCB Design (KiCad): Designed multi-sheet schematics and compact PCB layouts adhering to DFM standards, incorporating power distribution networks, MCU breakouts, and communication transceiver interfaces.
- MCU Peripheral Drivers & Firmware (C/C++ & Python): Implemented robust drivers for hardware communication buses (I2C, SPI, UART) across ESP32, STM32, and Arduino platforms.

TECHNICAL SKILLS

Hardware & PCB Design	KiCad (Schematic & Layout), Circuit Prototyping, Sensor Interfacing, Breadboarding, Hardware Debugging
Embedded & IoT	ESP32, STM32, Arduino, Raspberry Pi, Communication Protocols (UART, SPI, I2C, LoRa), IoT Node Architecture
Programming Languages	C, C++, Python
Software Tools & Platforms	MATLAB & Simulink, Git, Linux / Bash, Database Connectivity, Basic Machine Learning / Data Processing
Core Competencies	Project Management, Critical Thinking, Analytical Problem Solving, Multidisciplinary Collaboration

CERTIFICATES

- From Schematic to PCB: Layout Fundamentals — Coursera / Maheen Rasheed (May 2026)
- Introduction to the Internet of Things and Embedded Systems — Coursera
- MATLAB Onramp — MathWorks
- LIFT UP - Sanayi Odaklı Lisans Bitirme Projeleri Konferansı
- English Language School Program (B2 Certificate) — English Path Dubai (August 2025)

LANGUAGES

- Turkish: Native
- English: Upper-Intermediate / Professional Working Proficiency (B2)